

BS SE/CS F23

Operating Systems

Assignment 1

Max Marks:15+10+10=25

Deadline: 4 May 2026

Question 1:

The following processes are being scheduled by the system. In addition to the processes listed below, the system also has an idle task (which consumes no CPU resources and is identified as idle). This task has priority 0 and is scheduled whenever the system has no other available processes to run.

- A) a pre-emptive round robin scheduling algorithm (ignore process priorities for this part). The length of a time quantum is 10 units.
- B) A pre-emptive priority scheduling algorithm. Each process is assigned a numerical priority, with a higher number indicating a higher priority.

Thread	Priority	Burst	Arrival
P1	40	20	0
P2	30	25	25
P3	30	25	30
P4	35	15	60
P5	5	10	100
P6	10	10	105

For both parts (part A and part B), do the following tasks:

- 1) Show the scheduling order of the processes using a Gantt chart.
- 2) What is the turnaround time for each process?
- 3) What is the waiting time for each process?
- 4) What is the CPU utilization rate?
- 5) Compare both scheduling algorithms in terms of average turn-around time.
- 6) Compare both scheduling algorithms in terms of average waiting time.

Question 2:

Write C/C++ code to implement round robin scheduling algorithm. Design the format of your input text file and explain in readme file. Process ID, arrival time, burst time and quantum should be input through text file. Each process's waiting time, response time and turn-around time should be displayed on screen and recorded to an output file too. Gantt chart should be displayed on screen only. Algorithm statistics like average response time, average turn-around time and average waiting time should also be displayed on screen.

Question 3:

Write (C/C++ or Python or Java) code to implement a multi-level feedback queue (MLFQ) scheduling algorithm with given specifications.

Queue No.	Quantum	Algorithm	Queue Priority
Q1	5 msec	RR	1
Q2	8 msec	RR	2
Q3	-	SRTF	3

- 1) Input details of n number of processes from a text file. Process details will include process ID, arrival time, burst time and priority value. Use appropriate data structures to save process details after reading from file. Then use appropriate data structures to implement three ready queues and generate time line (Gantt Chart).